Can excitonic-insulator switches beat CMOS? A benchmarked feasibility screen across the BEC–BCS crossover

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Exciton condensates and excitonic insulators have been proposed as switches that operate below the $\ln(10)k_BT/e \approx 60$ mV/decade thermionic limit of complementary metal-oxide-semiconductor (CMOS) transistors, which would reduce the energy cost of computing. We test this premise with a set of mean-field models in which every ingredient is first benchmarked against a published result: a two-dimensional (2D) electron–hole gap equation across the BEC–BCS crossover; self-consistent screening in the superfluid state, validated against diffusion quantum Monte Carlo (DQMC); massless Dirac bands for graphene double layers; a quasi-one-dimensional three-chain model of Ta_2NiSe_5 with exciton–phonon coupling; Gaussian order-parameter fluctuations; gate electrostatics; and Abrikosov–Gor'kov disorder. Screening together with the Nelson–Kosterlitz bound gives a ceiling $T_c \leq \mathrm{Ry}^*/4r_{s,c}^2$, where $r_{s,c}(d/a_B)$ is the density at which the superfluid appears. No hBN-encapsulated double layer reaches 300 K. For monolayer-graphene double layers at the operating point of the bilayer pseudospin field-effect transistor, the gap vanishes under both normal-state and superfluid-state screening. In Ta_2NiSe_5 the ordered state is ≈70% excitonic but requires the lattice, and its discrete symmetry breaking precludes superfluidity. The only sub-CMOS switching found, an abrupt mean-field gap onset, does not survive Gaussian fluctuations: the swing becomes 100–470 mV/decade at 300–350 K, rising to ≥285 mV/decade once gate electrostatics are included. Switching energy at equal geometry is therefore 2–45 times higher than ideal CMOS. A machine-learning surrogate trained on the benchmarked physics finds no buildable design above ≈120 K with hBN dielectrics. These results identify 2D phase stiffness under screening, rather than pairing, as the bottleneck for excitonic low-power logic.

I. INTRODUCTION

Electricity use by data centres is projected to roughly double between 2024 and 2030, with artificial intelligence as the main driver[1]. A fundamental limit on the energy per logic operation in CMOS is the subthreshold swing $S \geq \ln(10)k_BT/e$, which sets the minimum gate voltage per decade of current, 60 mV/decade at room temperature[2]. Devices that sidestep this "Boltzmann tyranny" are actively sought.

Excitonic systems have been proposed in this context. An excitonic insulator (EI) forms when the single-particle gap E_g is smaller than the exciton binding energy E_b, and the ground state is a BCS-like condensate of electron–hole pairs[3,4]. In spatially separated bilayers[5] the condensate supports counterflow superfluidity. On this basis the bilayer pseudospin field-effect transistor (BiSFET) was proposed, with a simulated switching energy of 0.008 aJ per clock cycle, compared with ∼5 aJ for end-of-roadmap CMOS[6]. Its key assumption is a room-temperature condensate in graphene double layers, based on an unscreened mean-field estimate $T \sim 0.1E_F$[7]. Including carrier screening lowers this estimate by many orders of magnitude[8]. Experimentally, signatures of excitonic order exist in Ta_2NiSe_5[9–11], 1*T*-$TiSe_2$[12], monolayer WTe_2[13,14], TMD double layers[15,16], InAs/GaSb[17] and graphene double bilayers in the quantum Hall regime[18,19]. Room-temperature excitonic transistors exist, but they operate without a condensate[20].

This work asks a narrow question. Given the pairing physics as understood in the literature, can an excitonic switch beat CMOS at chip temperatures (300–350 K)? The approach is a chain of models in which each ingredient reproduces a published number before it is used (Table I), so the conclusion does not rest on unvalidated approximations. All code is openly available[21].

II. MODELS AND METHODS

A. 2D continuum gap equation

Electrons and holes with reduced mass m_r pair through $V(q) = 2\pi e^2 e^{-qd}/[\epsilon q(1 + r_0 q)]$, with layer separation d, background dielectric constant ϵ and Rytova–Keldysh length r_0[22]. The gap equation is

$$\Delta_k = \sum_{k'} V_{kk'} \frac{\Delta_{k'}}{2E_{k'}} \tanh \frac{E_{k'}}{2k_BT}, \qquad E_k = \sqrt{\xi_k^2 + \Delta_k^2}, \qquad \xi_k = \frac{E_g}{2} + \frac{\hbar^2 k^2}{4m_r}, \tag{1}$$

where $\mu_{\mathrm{eff}} = -E_g/2$ plays the role of the chemical potential in the BEC–BCS crossover[23]. The radial Nyström discretisation uses an analytic angular integral (complete elliptic integrals) and cell-averaging of the integrable logarithmic singularity. In 2D the mean-field T^* is only a pairing scale on the BEC side[23]. The condensation temperature is bounded by the Nelson–Kosterlitz jump[24], $k_BT_{\mathrm{BKT}} \leq \pi\hbar^2 n_s/2M$ with $M = m_e + m_h$ and $n_s \leq n$.

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B. Screening in the superfluid state

At fixed pair density n [effective units a_B^* and Ry*; $r_s = (\pi n)^{-1/2}/a_B^*$; one species per layer[25]], the pairing interaction is screened by the carriers of the paired state [superfluid-state (SS) screening][26]:

$$V_q^{\mathrm{SS}} = \frac{v_q e^{-qd}}{1 + 2v_q\Pi(q) + [v_q\Pi(q)]^2(1 - e^{-2qd})}, \qquad (2)$$

with $v_q = 2\pi e^2/\epsilon q$, and $\Pi = \Pi_n + \Pi_a$ is built from BCS coherence factors, $\Pi(q) = \sum_k (u_k v_{k'} - v_k u_{k'})^2/(E_k + E_{k'})$ with $k' = k+q$. It reduces to the static 2D Lindhard function for $\Delta \to 0$ and vanishes as $q \to 0$ in the superfluid. Iteration starts from the unscreened (US) coherent state. Neilson *et al.* showed that this SS approach, unlike US or normal-state (NS) screening, reproduces the DQMC condensate fraction[26].

C. Dirac bands

For graphene double layers we use the two-band Dirac model[7], $\xi_k = \hbar v k - E_F$, coupling $\alpha = e^2/\epsilon\hbar v$, chirality factor $(1 + \cos\theta)/2 = [(k + k')^2 - q^2]/4kk'$ and four spin–valley copies. Three screening treatments are compared: US[7]; NS random-phase approximation[8]; and SS screening as in Eq. (2), with the intraband polarisability from coherence factors and the interband part kept at its normal-state value.

D. Quasi-1D chains: Ta_2NiSe_5

We use the three-chain Hubbard model of Kaneko *et al.*[27]. It has two Ta $5d$ conduction chains and one Ni–Se valence chain, with $t_c = -0.8$, $t_f = 0.4$ and $D = 0.2$ eV fitted to density-functional theory. It includes on-site U, interchain $V = U/4$, and a uniform shear phonon with $\lambda = g^2/\omega$. Since the mean-field equations of Ref. 27 are given in supplementary material, we derive them from its Hamiltonian: Hartree terms for U and V, Fock terms on both c–f bonds, a static $q = 0$ phonon, and opposite order on the two Ta chains.

E. Fluctuations, gate and disorder

Gaussian fluctuations are included through a Ginzburg–Landau expansion of the Gibbs–Bogoliubov variational free energy along the mean-field pair shape, with stiffness $\hbar^2/2M$. The quartic term is treated in the self-consistent Hartree (large-N) approximation[28,29] with Bose occupation of the pair modes, $\langle|\psi|^2\rangle = \sum_Q n_B(a + b\langle|\psi|^2\rangle + cQ^2)$. Gaussian fluctuations about the normal state are equivalent to the Nozières–Schmitt-Rink scheme[23,30]. The fluctuation-broadened gap follows the pseudogap form $\Delta^2 = \Delta_{\mathrm{sc}}^2 + \Delta_{\mathrm{pg}}^2$[31]. Gate electrostatics follow Ref. 32: $e\,dV = (C + d\mu_x/dn)\,dn$ with $C = 4\pi e^2 d_s/\epsilon$. A thermal compressibility term k_BT/n is added because the pair gas is non-degenerate at 300 K. Disorder enters through the Abrikosov–Gor'kov equation as applied to bilayers[33].

F. Device metrics

For a gap-switching transistor with activated conduction $G \propto e^{-E_{\mathrm{gap}}/2k_BT}$, the swing is $S = 2\ln(10)k_BT/|dE_{\mathrm{gap}}/d(eV)|$, which is compared with $S_{\mathrm{CMOS}} = \ln(10)k_BT/e$. The mean-field overestimate of T_c by $(4e)^{1/3} \approx 2.2$[23,34] is applied as a robustness correction when deciding whether pairing survives at the operating temperature. For the BiSFET, the criterion is a 2D condensate at the operating temperature. Switching energy at equal geometry scales as $(S_{\mathrm{EI}}/S_{\mathrm{CMOS}})^2$. As a sanity check, C_gWLV^2 for the BiSFET geometry gives 0.0044 aJ, against the SPICE value of 0.008 aJ[6].

III. RESULTS

A. Crossover and the pairing–condensation gap

Figure 1 shows the 2D crossover for $MoSe_2$/hBN/WSe_2-like parameters ($m_r = 0.22$, $\epsilon = 4.5$, $d = 1.6$ nm). On the BCS side, Δ_k peaks at the Fermi wavevector and v_k^2 has a Fermi edge. On the BEC side, both are smooth. With the unscreened interaction, the mean-field pairing scale T^* exceeds 350 K over most of the EI window, but the BKT bound keeps the condensation temperature below ≈200 K, and at only tens of kelvin near the crossover. Screening (Sec. III B) removes the high-density end. This separation of T^* and T_c on the BEC side mirrors the cold-atom phase diagram[23]. It is the first sign that the bottleneck is phase stiffness, not pairing.

B. Screening ceiling

With superfluid-state screening, a large-gap superfluid exists only for $r_s > r_{s,c}(d/a_B)$, in agreement with diffusion quantum Monte Carlo [Fig. 2(a)]. Below $r_{s,c}$, screening suppresses the gap exponentially[26]. Combining this with $n_s \le n$ gives

$$k_BT_c \le \frac{\mathrm{Ry}^*}{4r_{s,c}^2}\,\frac{4m_r}{m_e + m_h}, \qquad (3)$$

where equal masses is the most favourable case. With the $MoSe_2/WSe_2$ parameters of Ref. 32 ($m_r \approx 0.2$, $\epsilon \approx 5$, $d = a_B \approx 1.3$ nm) the bound is 57 K. With the representative values used in Figs. 1 and 4 ($m_r = 0.22$, $\epsilon = 4.5$, $d = 1.6$ nm) it is 49 K, and at the thinnest barrier, $d = 0.7$ nm, it is 112–121 K. Across $0.05 \le m_r \le 0.6$,

TABLE I. Benchmarks reproduced before each ingredient is used; all rows are regenerated by `run_benchmarks.py`[21]. Literature values read from figures are approximate ($\approx \pm 0.03$). Two Ta_2NiSe_5 rows are documented deviations (see text).

ingredient	quantity	literature	this work
2D gap equation	2D exciton $E_b = 4\,\mathrm{Ry}^*$	435.4 meV (exact)	432.9 meV
	EI exists only for $E_g < E_b$	Ref. 4	yes
crossover mean field	3D unitarity Δ/E_F, μ/E_F	0.68, 0.59[23]	0.686, 0.591
	BCS-limit T_c	$(8\gamma/\pi e^2)E_F e^{-\pi/2k_F\lvert a\rvert}$[23]	within 1%
fixed-density bilayer	$\Delta_{\max}$ at $d = a_B$, $r_s = 1.5, 3, 20$	0.66, 0.68, 0.18 Ry*[25]	0.659, 0.700, 0.173
SS screening	superfluid onset $r_{s,c}$, $d = a_B$	2.5 (DQMC), ≈ 3 (SS)[26]	2.38
	onset, $d = 0.7a_B$	2.1 (DQMC)[26]	1.97
Dirac, NS screening	$V_{\mathrm{us}}(p_F)/V(p_F)$, $\alpha = 1$	≈ 9[8]	8.99
	$\lambda_c^{\max}$	$1/16$[8]	0.0624
	$d\lambda/d\ln(1/T)$ vs. λ_c ($\alpha = 1$)	equal[8]	0.0546 vs. 0.0567
Ta_2NiSe_5 chains	distortion onset λ_c	0.125 eV[27]	0.125 eV
	EI onset V_c	0.6 eV[27]	0.69 eV (+15%)
	$k_B T_c$ ($V = 0.55$, $\lambda = 0.04$ eV)	0.0783 eV[27]	0.0564 eV (−28%)
gate	interlayer E_b, $d = a_B$	0.77 Ry* (linear fit)[32]	0.866 Ry* (exact)
disorder	$dT_c/d\delta$; critical δ	$-\pi/4$; $0.88T_{c0}$[33]	-0.7855; $0.8819T_{c0}$
energy	BiSFET switching energy	0.008 aJ[6]	0.0044 aJ (CV^2)

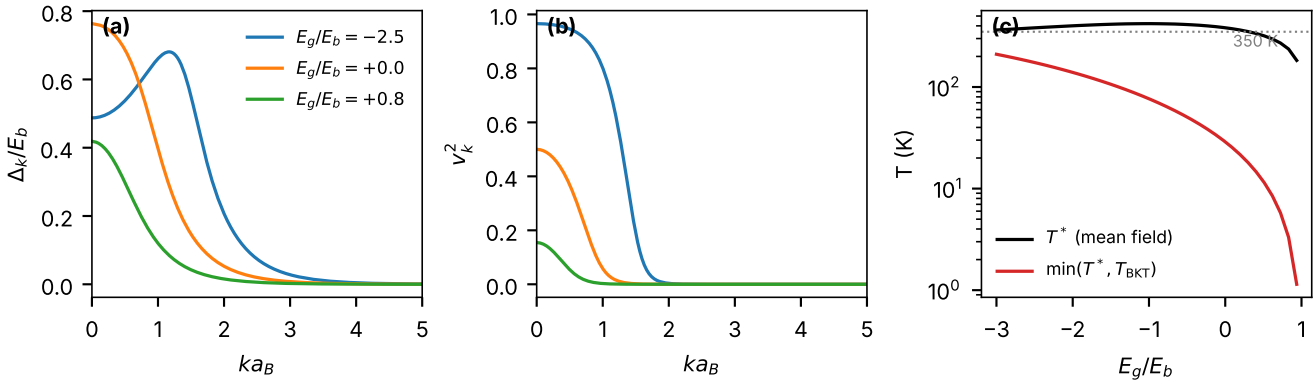


FIG. 1. BEC–BCS crossover in a 2D e–h double layer with the unscreened interaction ($m_r = 0.22$, $\epsilon = 4.5$, $d = 1.6$ nm; representative $MoSe_2$/hBN/WSe_2 values). (a) Gap function, (b) pair occupation, (c) mean-field pairing scale T^* and condensation estimate $\min(T^*, T_{\mathrm{BKT}})$ versus E_g/E_b.

$2 \le \epsilon \le 15$ and $0.6 \le d \le 3$ nm, no point reaches 350 K, and none with $\epsilon \ge 4.5$ reaches 300 K [Fig. 2(b)]. The maximum is 328 K, reached at $\epsilon = 2$. This result is unchanged when $r_{s,c}$ is extrapolated linearly below the smallest computed separation, $d/a_B = 0.3$. Valley and spin degeneracy add screening channels and lower the bound further.

C. Graphene double layers: the BiSFET operating point

At the BiSFET density ($n = 4.9 \times 10^{12}$ cm^{-2}, $E_F = 258$ meV), with $d = 1$ nm and $\epsilon = 3.9$, the unscreened Dirac gap is $\Delta = 31$ meV ($\approx 0.12E_F$), consistent with Ref. 7. This corresponds to $T_{\mathrm{KT}} \le E_F/8 \approx 374$ K. With normal-state screening the coupling is $\lambda_c < 1/16$ and $T_c \sim 10^{-7}E_F$[8]. With superfluid-state screening the gap collapses within two iterations. Even at $\epsilon = 1$ and $d = 0.3$ nm it converges to 0.087 meV. Monolayer-graphene double layers therefore do not condense at useful temperatures under either screening treatment validated in the literature. This agrees with the conclusion of Ref. 26 for monolayer graphene at 1 nm separation.

D. Ta_2NiSe_5

In the three-chain model (Fig. 3) the order is cooperative. At $V = 0.55$ eV, below the purely excitonic threshold, order appears only for $0.02 < \lambda < 0.04$ eV

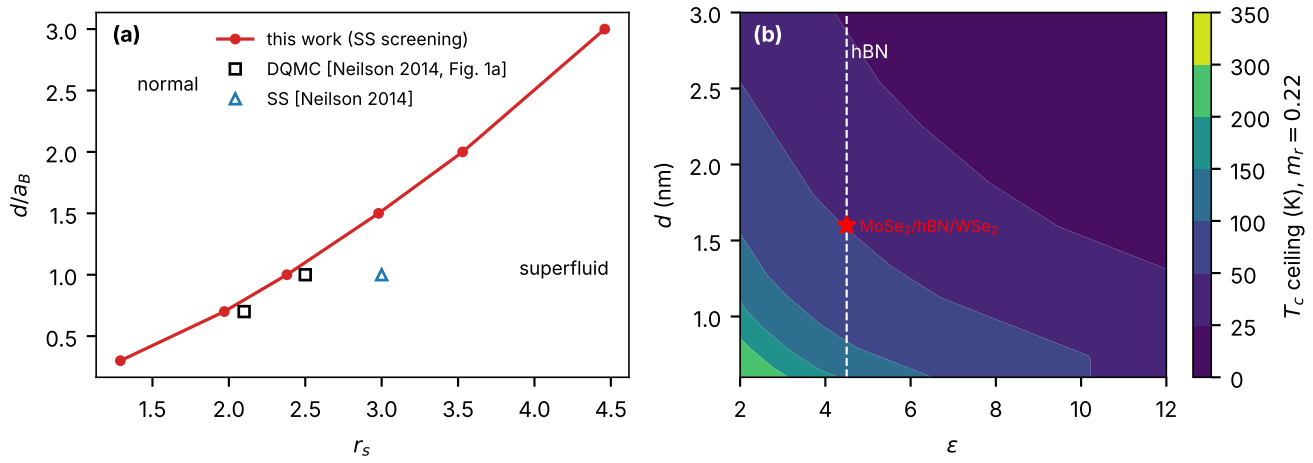


FIG. 2. (a) Superfluid onset $r_{s,c}$ versus layer separation with superfluid-state screening, compared with DQMC and mean-field SS points read from Ref. 26. (b) Screening ceiling, Eq. (3), for $m_r = 0.22$. The white contour marks 300 K and the dashed line marks hBN ($\epsilon = 4.5$).

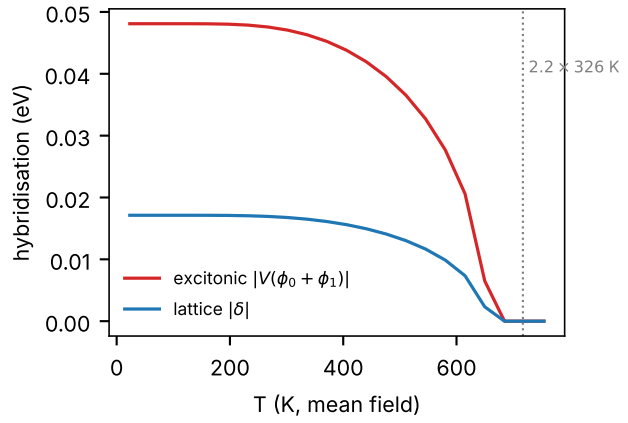


FIG. 3. Ta_2NiSe_5 three-chain model[27] at $V = 0.55$ eV, $\lambda = 0.04$ eV: excitonic and lattice contributions to the c–f hybridisation versus mean-field temperature.

or larger. For $\lambda = 0.04$–0.05 eV, 69–74% of the c–f hybridisation is excitonic. A purely structural instability would need $\lambda \geq 0.125$ eV, 2.5–6 times the estimate $\lambda \approx 0.02$–0.05 eV inferred from the measured distortion[27]. The mean-field T_c divided by 2.2 gives 298–377 K, which brackets the measured 326 K[10]. Symmetry analysis based on first-principles Wannier models shows that the transition breaks only discrete symmetries and that there is no excitonic superfluidity[35]. The structural interpretation of Ref. 36 is compatible with our finding that the lattice decides whether order sets in. For devices, Ta_2NiSe_5 cannot host a counterflow condensate, and as a bulk crystal its gap can be tuned only by pressure or strain.

E. Switching steepness and energy

The only sub-CMOS swing found in mean field comes from an abrupt gap onset where $T^*(E_g)$ crosses the operating temperature: 60.4 mV/decade at 350 K for $MoSe_2$/hBN/WSe_2 and 38.6 mV/decade at 300 K for WTe_2 (Fig. 4). Gaussian fluctuations replace this onset with a smooth pseudogap crossover, giving $S = 102$–140 and 137–280 mV/decade respectively across a factor-of-four range of momentum cut-off. The Ginzburg–Landau quartic reproduces the mean-field amplitude to within 4%. A real gate makes the situation worse. At 300 K, interlayer charging plus the thermal compressibility of the pair gas give $S \geq 285$ mV/decade. A negative exchange compressibility[32] alone would give 49 mV/decade, but this gain does not survive at non-degenerate densities. At equal geometry, the switching energy is 2.2–4.1 times ($MoSe_2$/hBN/WSe_2) and 4.5–45 times (WTe_2) that of ideal CMOS at 350 K. Disorder is not limiting at room temperature: preserving $T_{c0} = 350$ K requires only a pair-breaking time $\tau > 12$ fs[33].

F. Machine-learning screen

A bootstrap ensemble of gradient-boosted trees[37] was trained on 3000 solver labels $T_c = \min(T^*/2.2, T_{\mathrm{BKT}}, \text{Eq. (3)})$, with $0.01 \leq m_r \leq 1$, mass ratio 0.2–5, $1.5 \leq \epsilon \leq 15$ and $0.3 \leq d \leq 10$ nm. On held-out data it reaches $R^2 = 0.87$ with a mean absolute error of 1.3 K, and 95% of points fall within 2σ. Inverse design over 2×10^5 candidates, followed by verification with the full solver, gives at most ≈ 117 K for $\epsilon \geq 4.5$ and ≈ 233 K for $\epsilon \approx 1.6$. One label of 3000 exceeds 350 K, at $\epsilon = 1.7$, $d = 0.58$ nm and $E_g = -1.4$ eV, which is not a realistic device.

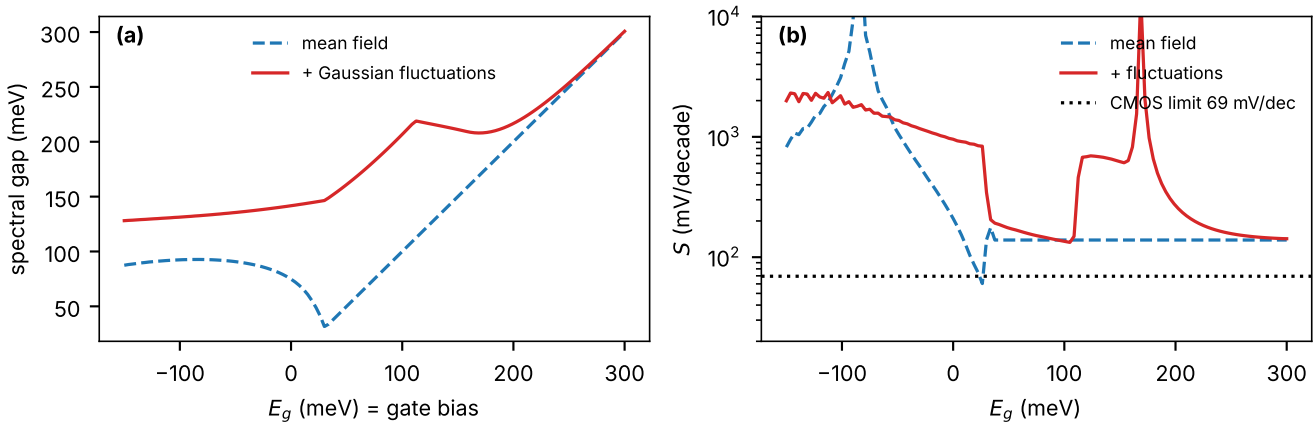


FIG. 4. Gap-switching transistor for $MoSe_2$/hBN/WSe_2 at 350 K. (a) Spectral gap versus E_g (ideal gate), in mean field and with Gaussian fluctuations. (b) Subthreshold swing, compared with the CMOS limit.

IV. DISCUSSION AND LIMITATIONS

The conclusion is robust because every ingredient that could raise T_c was treated optimistically: equal masses, $n_s = n$, an ideal gate in the bound, and single-species screening. The main limitations are: (i) static mean field with Gaussian fluctuations, without vortex dynamics or vertex corrections beyond those tested in Ref. 26; (ii) no intralayer exchange in the continuum model, which changes the sign of the exciton–exciton interaction compared with Ref. 32; we use both as brackets; (iii) a 15–28% deviation from two Ta_2NiSe_5 numbers of Ref. 27. Its mean-field equations are given in supplementary material that was not used here, and two alternative Fock decouplings bracket the published values; and (iv) no transport, contact or circuit modelling. Non-equilibrium (optical) switching in Ta_2NiSe_5 and cryogenic operation, where the CMOS limit falls to 15 mV/decade at 77 K, lie outside this screen.

V. CONCLUSION

Within benchmarked models of 2D parabolic, Dirac and quasi-one-dimensional excitonic systems, excitonic-insulator switches do not beat CMOS in subthreshold swing or switching energy at 300–350 K. Room-temperature pairing is possible, but screening-limited 2D condensation is not. The steep switching that appears in mean field is an artefact that Gaussian fluctuations remove. The design rule of Eq. (3) and the open code provide a quantitative test for future excitonic device proposals.

DATA AVAILABILITY

All code, data and figure scripts are openly available at `https://github.com/masyhur-n/excitonic-insulator-sim`[21] under the MIT license, and archived at Zenodo, doi:10.5281/zenodo.23684703[21].

AUTHOR DECLARATIONS

Conflict of interest. The author has no conflicts to disclose.

Author contributions. Nailah Masyhur: conceptualization, methodology, software, validation, formal analysis, writing (original draft, review and editing).

Ethics approval. Not applicable.

Use of AI tools. Code development, literature retrieval and drafting were assisted by an AI coding assistant (Claude, Anthropic). The author directed the study, verified every result against the published benchmarks listed in Table I, and takes full responsibility for the content.

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